

Electric Vehicles

Challenges and Barriers: A Review

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Introduction

- Road transport: 70% of transport-related CO₂ emissions
- EVs reduce CO₂ by 60 million tonnes globally (2023)
- 40% annual growth in EV adoption worldwide
- 30 million EVs on roads globally

Benefits of EVs

- Zero tailpipe emissions & lower maintenance costs
- 85-90% drivetrain efficiency vs 20-30% for ICE
- Annual electricity cost: US \$85 vs US \$1,117 for gasoline
- Reduced maintenance: US \$4,600 vs US \$9,200 for ICE vehicles

Global EV Market Status (2023)

1 in 4

New cars sold globally

90%

Norway market share

40%

China market share

EV Classification by Motor Type (Part 1)

- BEVs (Battery Electric Vehicles): Fully electric, no combustion engine, 160-500+ km range
- PHEVs (Plug-in Hybrid): Combines ICE + electric motor, 50+ km all-electric driving
- HEVs (Hybrid Electric): Regenerative braking, cannot be plugged in

EV Classification by Motor Type (Part 2)

- FCEVs (Fuel Cell Electric): Hydrogen fuel cells, 650 km range, zero emissions (only water vapor)
- EREVs (Extended-Range Electric): Small combustion engine recharges battery, 260 km electric + 130 km with range extender

Battery Technology Overview

- Lead-acid, Nickel-metal hydride, Lithium-ion
- Li-ion dominates due to superior energy density
- 65% increase in demand in 2022 (550 GWh globally)
- Cost reduction: \$1,200/kWh (2010) → \$150/kWh (2024)

Li-Ion Battery Advantages

- Energy density: 150-250 Wh/kg for extended range
- High power density for acceleration & fast charging
- Long lifespan: 1000+ charge cycles
- Low self-discharge rate vs lead-acid batteries

Major EV Challenges (Part 1)

- **Range Anxiety:** Most EVs: 200-350 km per charge
- **Charging Time:** Standard 4-8 hours, fast chargers reach 80% in 30 min
- **Battery Weight:** ~200 kg affects vehicle efficiency & handling
- **Environmental Impact:** Battery manufacturing concerns (recycling improving)

Major EV Challenges (Part 2)

- **High Upfront Cost:** 23% more expensive than ICE vehicles
- **Limited Infrastructure:** Insufficient charging networks, especially rural areas
- **Thermal Management:** Optimal range 25-40°C, degradation above 50°C
- **Material Scarcity:** Lithium, cobalt, nickel demand pressures

High Upfront Costs Breakdown

Average car: US \$8,008 | EV average: US \$58,900 (+23%)

- With US \$7,500 Tax Credit: Reduces to ~US \$51,000
- Tesla Model 3 base: US \$39,990 (price reduction trend)
- Annual electricity: US \$85 vs US \$1,117 for gasoline

Battery Thermal Management System (BTMS)

- Optimal temperature range: 25-40°C for Li-ion
- Temperatures >50°C drastically reduce lifespan
- Extreme temps cause rapid anode degradation
- Risk of thermal runaway at high temperatures
- BTMS critical for battery safety & reliability

Charging Infrastructure Gap

Current (2023): 2.7M points | Target (2030): 9M points

- Lowers range anxiety for long-distance travel
- Essential for urban dwellers without parking
- Smart charging prevents local grid overloads

CO₂ Emissions Reduction

- EVs reduced CO₂ by 60 million tonnes in 2023
- Zero tailpipe emissions eliminate local air pollution
- EV efficiency: 85-90% vs ICE: 20-30%
- Lifecycle emissions 70% lower than traditional cars with renewable grids

Emerging Technological Solutions (Part 1)

- **Solid-State Batteries:** Higher energy density, eliminate liquid electrolytes, improved safety
- **Ultra-Fast Charging:** Next-gen networks reduce charging time significantly
- **Vehicle-to-Grid (V2G):** EVs return power to grid, support renewable integration

Emerging Technological Solutions (Part 2)

- **AI & ML for Battery Health:** Predictive maintenance, aging prediction
- **Smart Charging Systems:** Manage energy demand, prevent grid overloads
- **Battery Recycling:** Extended battery life, circular economy, reduced mining

Solutions to Overcome Barriers

- Government incentives: tax credits, rebates, low-interest loans
- R&D investment for solid-state & sodium-ion batteries
- Expand public charging networks, especially highways & rural areas
- Promote battery recycling programs & circular economy

Future Outlook & Market Growth

- IEA projection: 60% of global vehicle sales by 2030 (up from 1% in 2022)
- Annual EV growth: 40% globally, 30 million on roads
- India's smart cities & sustainable transport goals rely on EVs
- Battery costs continue declining, improving affordability

Conclusion

Building Sustainable Transportation

Collaboration among governments, industries, consumers & academia

Overcoming barriers unlocks cleaner, greener mobility future